

Quiz 2

Subject: MLT 2025 Aug3: IIT M AN EXAM QIV4 Total Marks: 40.00 Questions: 19

QUESTIONS

Q1

MCQ

[0.00 marks]

THIS IS QUESTION PAPER FOR THE SUBJECT "**DIPLOMA LEVEL : MACHINE LEARNING TECHNIQUES (COMPUTER BASED EXAM)**" ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT? CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN. (IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

OPTIONS

A. YES ✓ Correct

B. NO

Consider n data points, with each feature $x_i \in \mathbb{R}^+$ and label $y_i \in \mathbb{R}$. Suppose $y_i \mid x_i \sim \mathcal{N}(\log(wx_i), 1)$, then find the ML estimate $\hat{w}_{MLE}$ of w , where $w > 0$.

OPTIONS

A.
$$\hat{w}_{MLE} = \exp \left(\frac{1}{n} \sum_{i=1}^n (y_i - \log(x_i)) \right)$$

✓ Correct

B.
$$\hat{w}_{MLE} = \exp \left(\frac{1}{n} \sum_{i=1}^n (x_i y_i - x_i \log(x_i)) \right)$$

C.
$$\hat{w}_{MLE} = \frac{1}{n} \sum_{i=1}^n (y_i - \log(x_i))$$

D.
$$\hat{w}_{MLE} = \exp \left(\frac{1}{n} \sum_{i=1}^n (x_i y_i - \log(x_i)) \right)$$

In the Bayesian modeling for linear regression, we assume the following:

- $y \mid x \sim \text{Normal}(w^T x, \sigma^2 I)$
- $w \sim \text{Normal}(0, \gamma^2 I)$

Under this setup, the maximum a posteriori (MAP) estimator of the weights $\hat{w}_{\text{MAP}}$ is given by:

$$\hat{w}_{\text{MAP}} = \arg \min_w \{ \|y - X^T w\|^2 + \lambda \|w\|^2 \}$$

Using the given information, find the expression for the regularization parameter λ .

OPTIONS

A.

$$\lambda = \sigma^2 \cdot \gamma^2$$

B.

$$\lambda = \frac{\gamma^2}{\sigma^2}$$

C.

$$\lambda = \frac{\sigma^2}{\gamma^2}$$

✓ Correct

D.

$$\lambda = \frac{1}{\sigma^2 + \gamma^2}$$

You are evaluating the performance of a linear regression model using different hyperparameter settings (learning rate and number of iterations). Below are the results:

Setting	Learning Rate	Iterations	Training Error	Validation Error
1	0.05	1000	150	160
2	0.10	800	145	150
3	0.15	500	140	190
4	0.25	350	170	180

Which hyperparameter setting would you choose based on the training and validation errors? Why?

OPTIONS

- A. **Setting 3**, because it has the lowest training error, indicating the best fit to the training data.
- B. **Setting 2**, because it has the lowest validation error and a relatively low training error, indicating a good generalization. ✓ **Correct**
- C. **Setting 1**, because it has the highest number of iterations and therefore likely to be more stable.
- D. **Setting 4**, because it uses the highest learning rate and converges the fastest.

Consider the linear regression model:

$$y = w^T x + \epsilon$$

where $\epsilon \sim \mathcal{N}(0, \sigma^2)$.

Note:

- $\hat{w}_{\text{MLE}}$: Maximum Likelihood Estimator (MLE) of w
- $\hat{w}_{\text{MAP}}$: Maximum A Posteriori (MAP) estimator of w
- $\mathbb{E}(\|\hat{w} - w\|^2)$: Mean Squared Error (MSE) between an estimator $\hat{w}$ and the true parameter w

Which of the following statements is/are true?

OPTIONS

- A. The estimator $\hat{w}_{\text{MLE}}$ always has a lower mean squared error (MSE) than $\hat{w}_{\text{MAP}}$ because it is unbiased.
- B. The estimator $\hat{w}_{\text{MAP}}$ always provides better estimates than $\hat{w}_{\text{MLE}}$ because it uses prior information..
- C. The estimator $\hat{w}_{\text{MAP}}$ can achieve lower MSE than $\hat{w}_{\text{MLE}}$ for some $\lambda > 0$, where $\lambda \in \mathbb{R}$.
✓ Correct
- D. The MSE of $\hat{w}_{\text{MLE}}$ and $\hat{w}_{\text{MAP}}$ is always equal when the sample size is small.

Consider a dataset $D = \{(x_1, y_1), \dots, (x_n, y_n)\}$. Let $w^* \in \mathbb{R}^d$ be the solution of the optimization problem $\min_{w \in \mathbb{R}^d} \|X^T w - y\|_2^2$, where $X \in \mathbb{R}^{d \times n}$ and $y \in \mathbb{R}^n$. Construct a new dataset D_{new} by duplicating all points in D and adding 10 new points that lie exactly on the line $y = x^T w^*$. Then, the solution w_{new}^* of the corresponding optimization problem is also w^* .

OPTIONS

- A. TRUE ✓ Correct
- B. FALSE

Q7 MCQ

[1.00 mark]

Consider the linear regression model $y = w^T x + \epsilon$. Assuming $\epsilon \sim \mathcal{N}(0, \sigma^2)$, maximizing the conditional log-likelihood of w is equivalent to minimizing the sum of squared errors $\|y - X^T w\|_2^2$.

OPTIONS

- A. TRUE ✓ **Correct**
- B. FALSE

Q8 MSQ

[2.00 marks]

Let $w^* \in \mathbb{R}^d$ be the solution of the optimization problem $\min_{w \in \mathbb{R}^d} \|X^T w - y\|_2^2$, where $X \in \mathbb{R}^{d \times n}$ and $y \in \mathbb{R}^n$. Then, which of the following statements is/are true?

OPTIONS

- A. $X^T w^*$ is the projection of the labels onto the subspace spanned by the features.
✓ **Correct**
- B. $X^T w^*$ is the projection of the labels onto the subspace spanned by the data points.
- C. w^* lies in the span of data points.
✓ **Correct**
- D. w^* lies in the span of features.

Which among the following are true?

OPTIONS

- A. In generative model, we are interested in how the features are generated.
✓ Correct
- B. In discriminative model, we are interested in how the features are generated.
- C. In discriminative model, we are interested in modelling $p(y|x)$, where y is the label and x is a feature.
✓ Correct
- D. In generative model, we are interested in modelling $p(y|x)$, where y is the label and x is a feature.
- E. Gaussian Naive Bayes is a discriminative model.

Perform a 3-NN regressor for the following dataset:

x_1	x_2	y
1	2	3.5
2	1	2.8
3	3	5.1
4	2	4.7
5	3	6.2
6	1	5.9

where, $x = [x_1 \ x_2]^T$ is a feature vector and y is the label. What will be the predicted value for the test data point $[3 \ 2]^T$? Enter the answer correct to one decimal place.

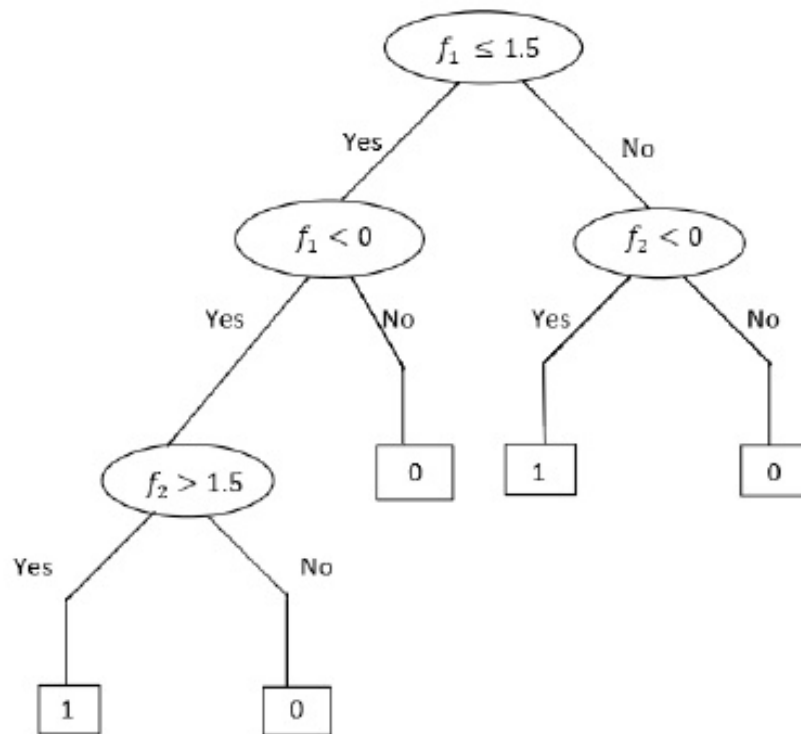
ANSWER

4.2 (Equal • Numeric)

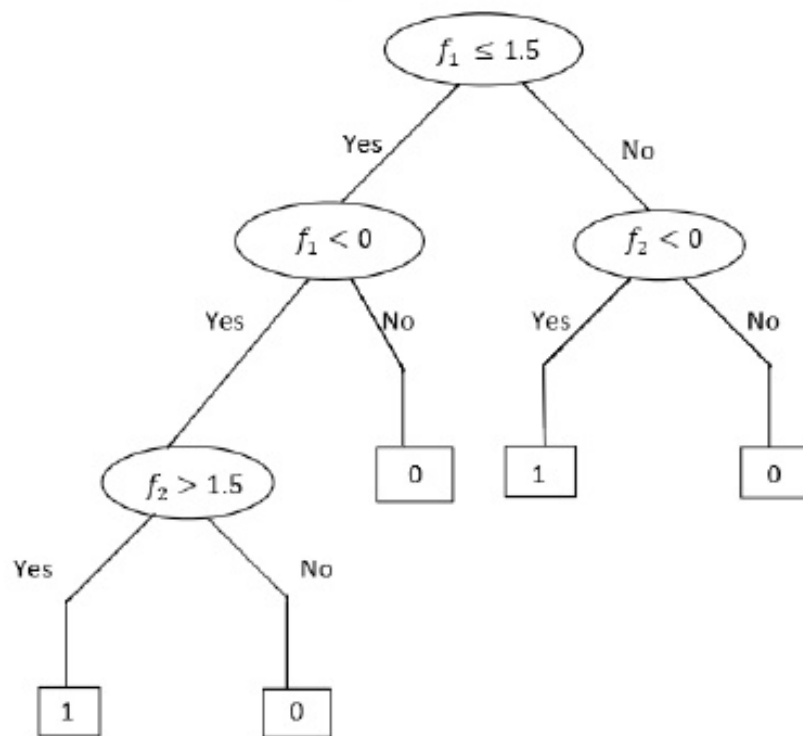
CONTEXT / PARAGRAPH

Based on the above data, answer the given subquestions.

Consider the following decision tree for a binary classification problem with two features f_1 and f_2 :



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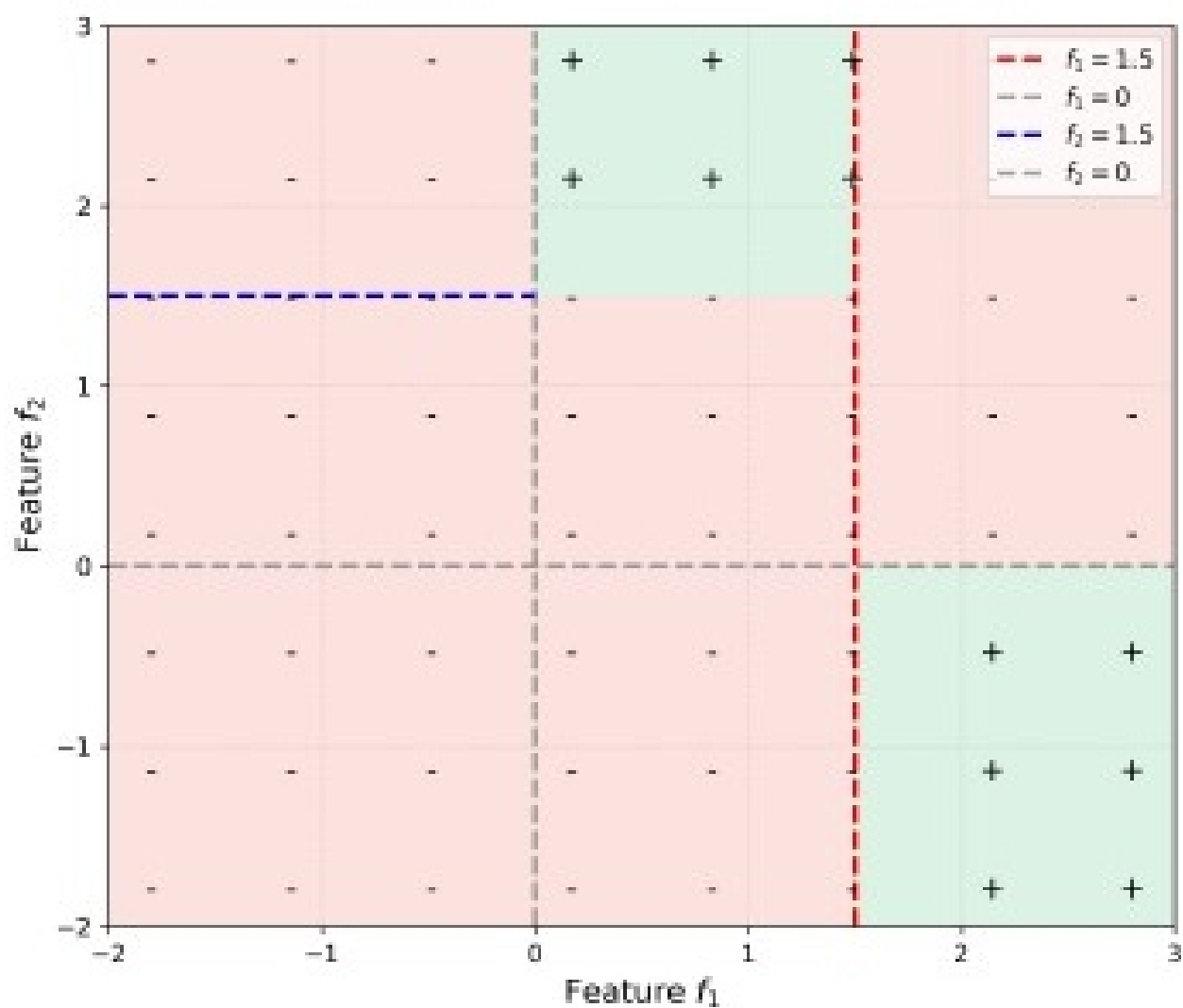


Which among the following represents the correct decision boundary for the decision tree?

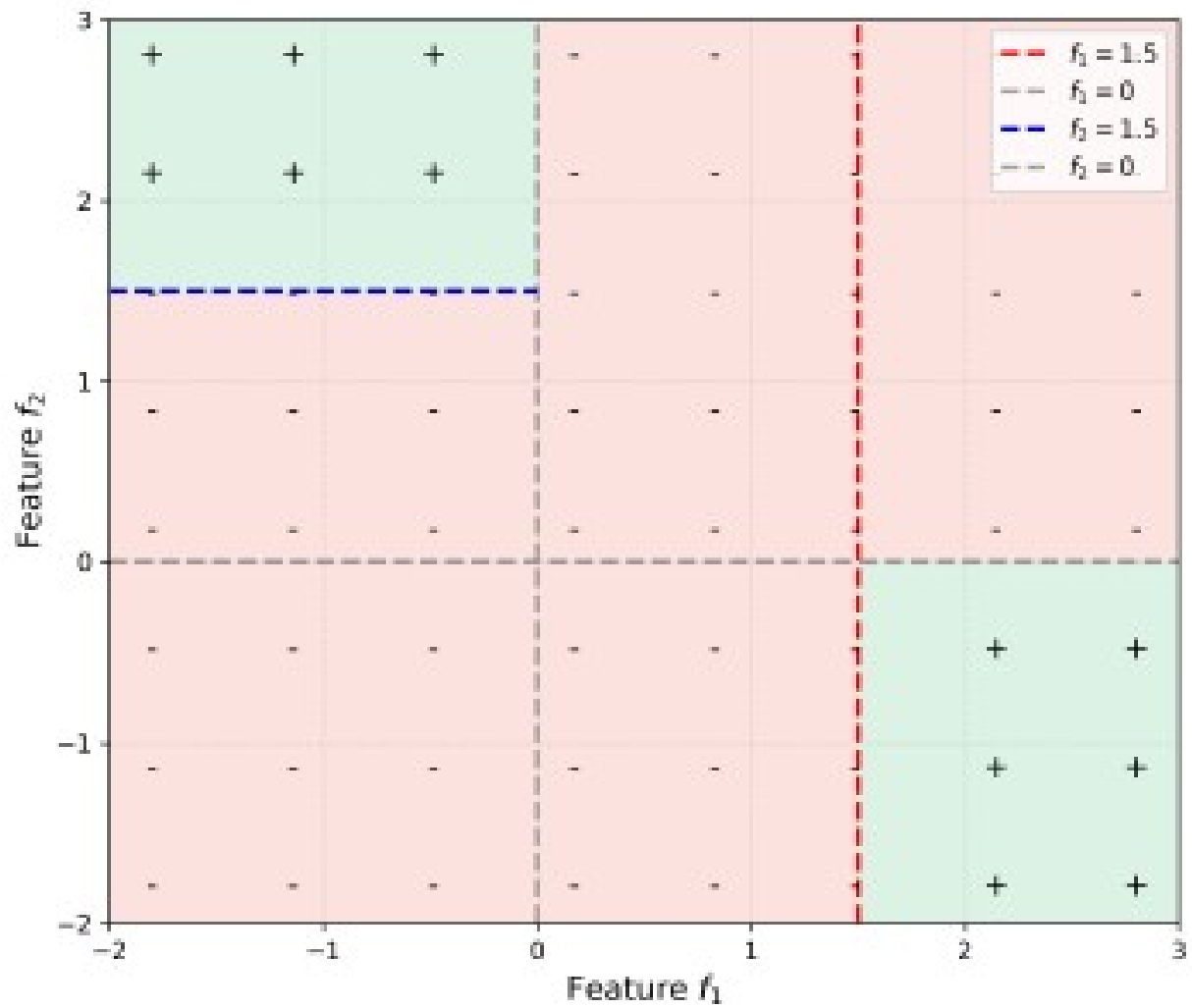
Note: Here + represents the label 1 and – represents the label 0.

OPTIONS

A.

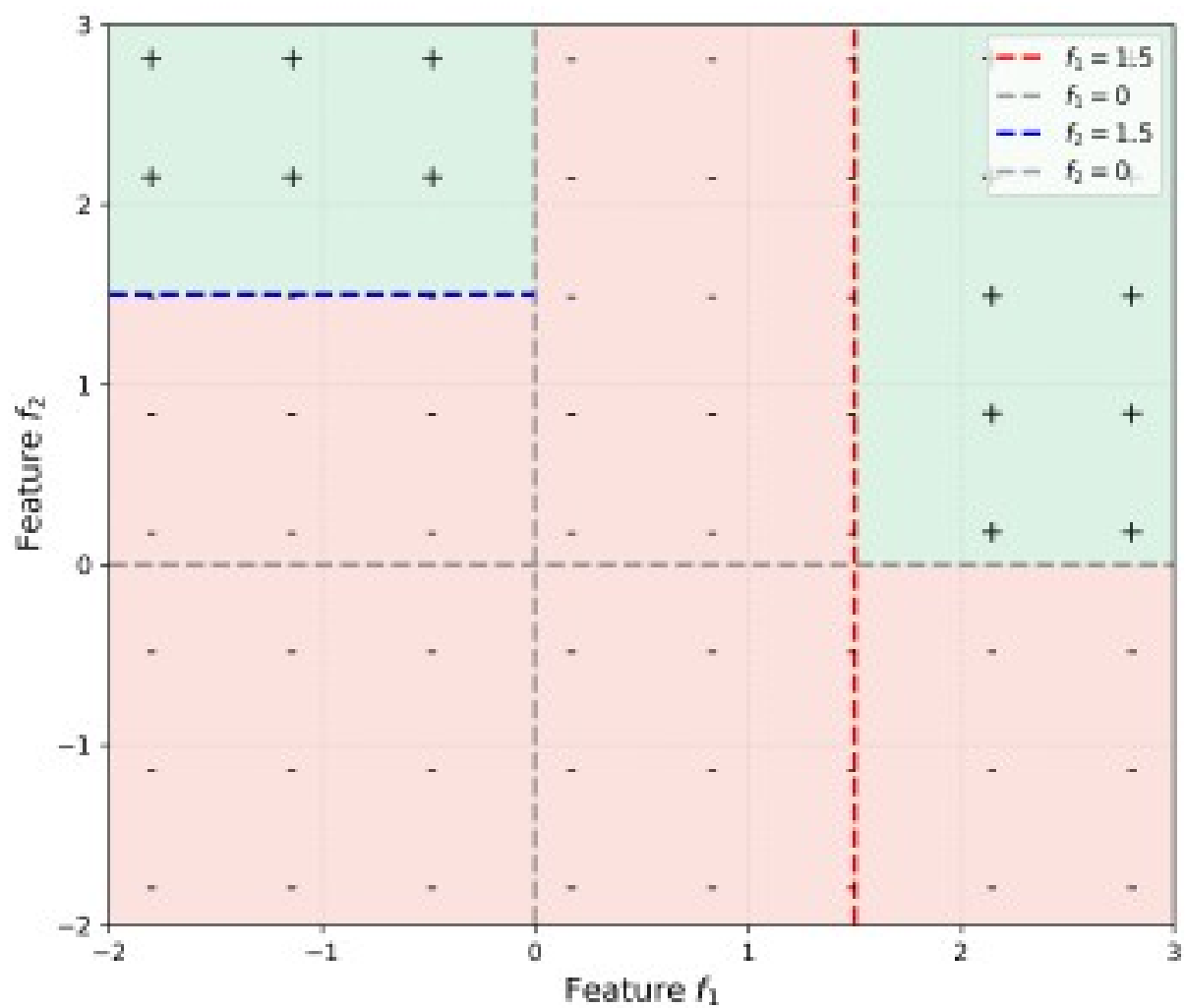


B.

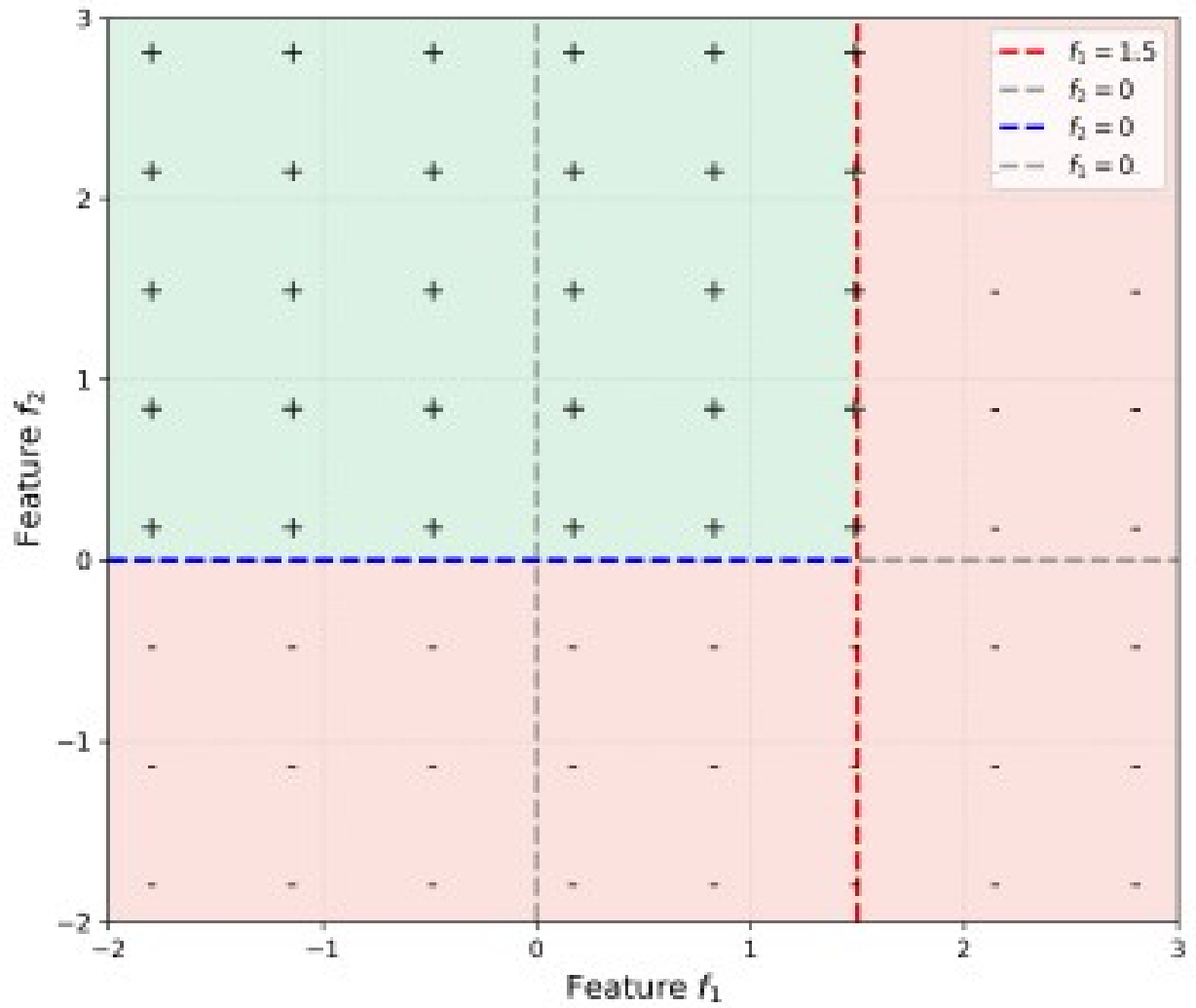


✓ Correct

C.



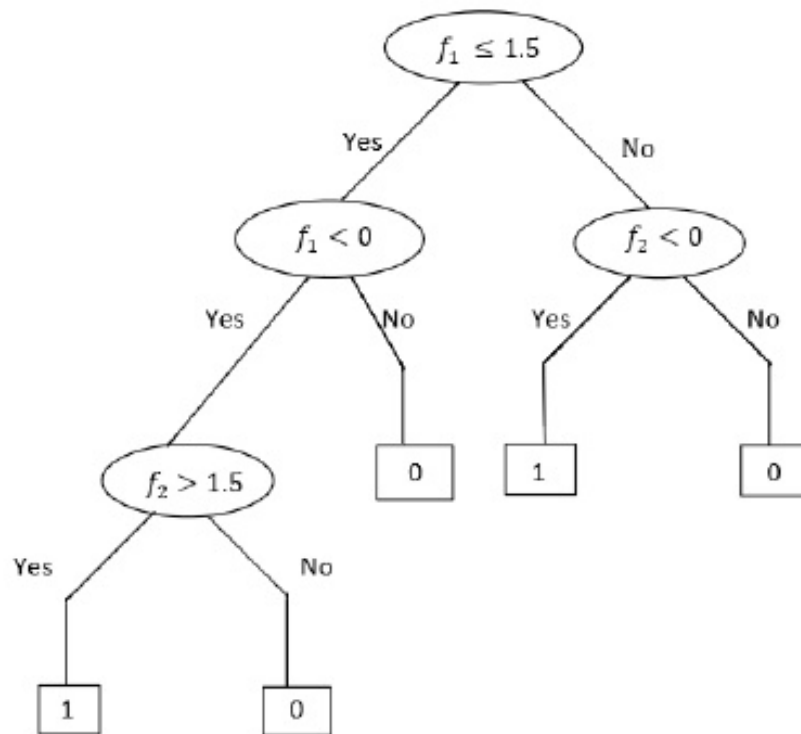
D.



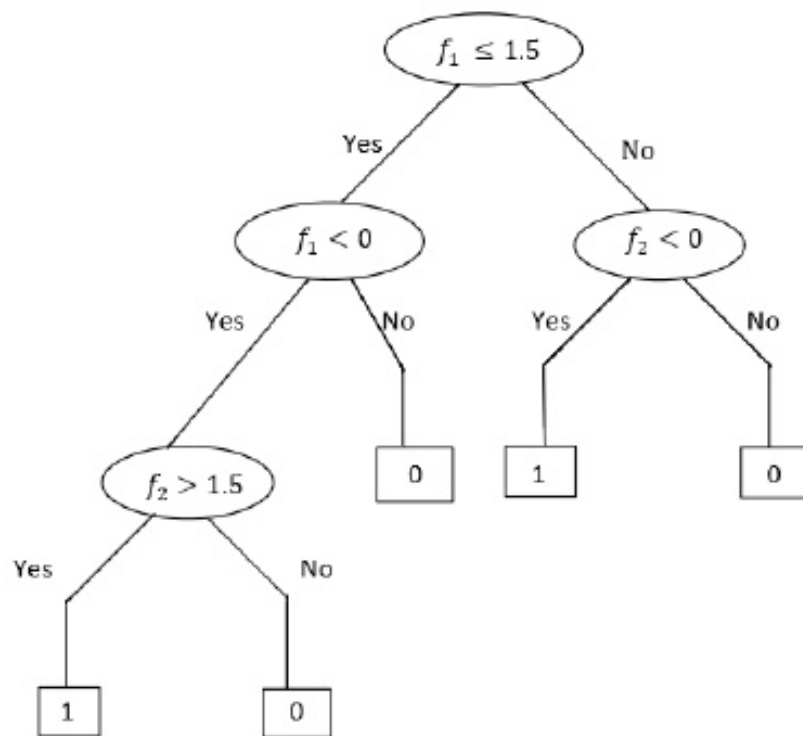
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What will be the prediction for the new datapoint (2, 2)?

CONTEXT / PARAGRAPH

Based on the above data, answer the given subquestions.

Consider a 2-class Naive Bayes classifier with first feature f_1 as binary and the second feature f_2 as Gaussian. Following are the information that are provided to us:

Estimate of $P(Y = 1), \hat{p} = 0.4$

Estimate of $P(f_1 = 1 \mid y = 0), \hat{p}_1^0 = 0.4$

Estimate of $P(f_1 = 1 \mid y = 1), \hat{p}_1^1 = 0.5$

$(f_2 \mid y = 0) \sim \text{Normal}(0, 1)$

$(f_2 \mid y = 1) \sim \text{Normal}(1, 1)$

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Estimate of $P(f_1 = 1 \mid y = 1), \hat{p}_1^1 = 0.5$

$(f_2 \mid y = 0) \sim \text{Normal}(0, 1)$

$(f_2 \mid y = 1) \sim \text{Normal}(1, 1)$

If a test point has label 0, what will be the probability that the point is

$x_{test} = [0 \ 0]^T$? Enter the answer correct to two decimal places.

ANSWER

0.22 - 0.26 (Range • Numeric)

CONTEXT / PARAGRAPH

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Estimate of $P(f_1 = 1 \mid y = 1), \hat{p}_1^1 = 0.5$

$(f_2 \mid y = 0) \sim \text{Normal}(0, 1)$

$(f_2 \mid y = 1) \sim \text{Normal}(1, 1)$

For $x_{test} = [0 \quad 1.5]^T$, what output would Naive Bayes model predict?

ANSWER

1 (Equal • Numeric)

CONTEXT / PARAGRAPH

Suppose you want to build a decision tree to help diagnose a certain disease based on patient data. You have a dataset of 1000 patients labeled as either “positive” or “negative”. Here, positive represents patients with the disease and negative represents no disease. One of the feature used to diagnose is headache.

Below is the breakdown of the patients reporting headache: Headache present: 300 positive, 50 negative Headache not present: 100 positive, 550 negative

Suppose you want to build a decision tree to help diagnose a certain disease based on patient data. You have a dataset of 1000 patients labeled as either “positive” or “negative”. Here, positive represents patients with the disease and negative represents no disease. One of the feature used to diagnose is headache.

Below is the breakdown of the patients reporting headache: Headache present: 300 positive, 50 negative Headache not present: 100 positive, 550 negative

Based on the above data, answer the given subquestions.

Use the following values if required:

- $\log_2(0.4) = -1.32$
- $\log_2(0.6) = -0.74$
- $\log_2(0.86) = -0.22$
- $\log_2(0.14) = -2.84$
- $\log_2(0.15) = -2.74$
- $\log_2(0.85) = -0.23$

Note: Use $\log_2$ for the entropy calculations.

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What is the entropy of the root node? Enter the answer correct to two decimal places.

ANSWER

0.95 - 0.99 (Range • Numeric)

CONTEXT / PARAGRAPH

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What is the information gain for choosing the feature “headache” for splitting at the root node?
Enter the answer correct to two decimal places.

ANSWER

0.34 – 0.38 (Range • Numeric)

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